

LABORATORY MEDICINE PROGRAM

**University
Health
Network**

DEPARTMENT OF LABORATORY HEMATOLOGY

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Blood and Bone Marrow Consultation Report

Patient Name:	Redacted	Accession #:	B22-6528
MRN:	Redacted	Service:	Redacted
DOB:	Redacted	Visit #:	Redacted
		Collected:	Redacted
		Received:	Redacted
Gender:	Redacted	Location:	Redacted
HCN:	Redacted	Facility:	Redacted
Ordering MD:	Redacted	Reported:	Redacted
Copy To:	Redacted		

Material Received

1. Blood for molecular diagnostics

Other Case Numbers

M22-08058

Diagnosis

See Genetics Report(s) Below.

Electronically Verified By:

zu6/2/2022

Clinical History

HBOP Panel

Comment: COPY

Molecular Diagnostics

Status: Signed Out

Date Reported: Redacted

Interpretation

Molecular Number: M#22.08058

RESULTS

Results: Variant(s) detected

- - Variant 1

Variant: Placeholder

Zygosity: Placeholder

Classification: Placeholder

- - Variant 2

Variant: Placeholder

Zygosity: Placeholder

Classification: Placeholder

Comments

The Placeholder and Placeholder variants are present in Placeholder in this patient.

Specimen tested: Placeholder

Pathogenic Variant Identified:

Placeholder

Germline pathogenic variants in Placeholder are associated with Placeholder including, but not limited to, breast cancer and colon cancer (PMID: 274859)

Based on the following evidence, Placeholder variant is classified as pathogenic: 1) this frameshift variant leads to a premature termination codon, which is predicted to result in a truncated or absent protein; 2) this variant is reported in association with an increased risk of developing breast cancer (PMID: 18172190, 24918820, 17428320); 3) this variant has been reported in families affected with Li-Fraumeni syndrome (PMID: 10617473, 11479205); 4) this variant is present in gnomAD (MAF = 0.21%, 591/280390 alleles, 1 homozygote) with a higher frequency reported in the European (Finnish) subpopulation (MAF = 0.87%, 219/25124 alleles); 5) this variant has been reported in ClinVar by 22 clinical laboratories, with conflicting classifications (20 pathogenic, 2 uncertain). Based on currently available information, this variant is classified as pathogenic according to the ACMG 2015 variant classification guidelines (PMID: 25741868). Although this variant is not expected to cause highly penetrant Mendelian disease, it is an established cancer risk factor.

Variant of Uncertain Significance Identified:

Placeholder

Based on the following evidence, the Placeholder variant is classified as a variant of uncertain significance: 1) this variant is present in gnomAD (MAF = 0.045%, 127/280624 alleles) with a higher frequency reported in the East Asian subpopulation (MAF = 0.42%, 84/19952 alleles); 2) this variant has been reported in the literature in individuals with familial breast/ovarian cancer, in patients with prostate cancer and in controls (PMID: 31472684, 30851065, 30287823, 29667044, 29506128, 29470806, 29338689, 29020732, 28961279, 27751358, 27621404, 27442652, 26787654, 25884806, 23960188, 21244692, 24390236, 21618645, 18484200); 3) this variant has been reported as a de novo occurrence in an individual with Li-Fraumeni-like syndrome (PMID: 27442652); 4) this variant has been reported to be associated with increased breast cancer risk (PMID: 21618645); 5) functional assays report conflicting effects of this variant on protein function (PMID: 30851065, 21618645); 6) this variant has been reported in ClinVar by 14 clinical laboratories as a variant of uncertain significance; 7) in silico algorithms provide inconsistent predictions with respect to the pathogenicity of this variant. Based on currently available information, this variant is classified as a variant of uncertain significance according to the ACMG 2015 variant classification guidelines (PMID: 25741868).

Genes(s)/Panel Tested:

Hereditary Breast/Ovarian/Prostate Cancer:

Placeholder

Methodology: Genomic DNA was analyzed using UHN Hereditary Panel version 4.0 to examine the exonic coding regions (+/- 10bp of the intron) of the listed genes, using the Illumina DNA Prep with enrichment hybrid capture followed by paired-end sequencing with the Illumina sequencing platform. Variant calls are generated using the UHN Clinical Laboratory Genetics custom bioinformatics pipeline with alignment to the reference genome sequence build GRCh37/hg19, and variants assessed using Alissa Interpret. For panels that include the EPCAM or GREM1 genes, only copy number variants are evaluated. Long-range PCR and Sanger sequencing were performed for confirmation of variants identified in PMS2 due to regions of high homology. Minimum acceptable coverage for

all reported genomic regions is $>25\times$. Exon level deletions or duplications were analyzed using DECON [PMID: 28459104] or using multiplex ligation-dependent probe amplification (MLPA) assays by MRC Holland. The reportable range is 20%-100% variant allele frequency. Test sensitivity for this assay is $>98\%$. Current methodology may not detect all variants or CNV changes present in the tested genes

Variant Class Reported: Variants were classified as pathogenic, likely pathogenic, uncertain, likely benign or benign according to 2015 ACMG guidelines (Richards et al 2015, PMID 25741868). Variants classified as likely benign or benign are not reported but are available upon request.

Regions with less than 25X read depth:

None.

A negative finding does not rule out the possibility of a hereditary predisposition to cancer. We recommend that patients be referred to genetic counseling to understand the implications of this result. Molecular testing is available for the patient's family once a mutation is identified in the proband.

The performance characteristics of this test were determined and verified by the Genome Diagnostics Laboratory at the University Health Network. This test has not been cleared or approved by the US FDA. FDA does not require this test to go through premarket FDA review. This test is used for clinical purposes; it should not be regarded as investigational or for research. This laboratory is certified under the Clinical Laboratory Improvement Amendments (CLIA) as qualified to perform high complexity clinical laboratory testing

Mock specimen